Fragmentation Calculator Algorithm

Step 1: Start

Step 2: Declare Variables p, BH, E, SL, BD, K, A, Q, X50

Step 3: Input E and carry over in-hole density(p), blasting diameter (BD), stemming length (SL), bench height (BH) from the previous surface blast design code.

Step 4: Determine Rock Factor(A)

Step 5: Calculate explosive mass per hole (Q)

Step 6: Calculate Mean Fragment Size (X50) using the Kuznetsov equation

Step 7: Display Mean Fragment Size (X50)

Step 8: END

Pseudocode

```
INPUT Relative Effective Energy of Explosive

FUNCTION FragmentationCalculator (ρ, BH, E, SL, BD, K, RockType)

    If RockType is "very soft":

        RETURN 6

    If RockType is "soft":

        RETURN 8

    If RockType is "medium":

        RETURN 10

    If RockType is "hard":

        RETURN 12

    Else:

        RETURN "Rock type is invalid"

Q ← ρ × (π × BD^2 / 4) × (BH − SL)

X50 ← A × K^(-0.8) × Q^ (1/6) × (116 / E)^(19/30)
```

DISPLAY X50

END FUNCTION

Results

The following figures show the flowcharts of the algorithms for the BlastSmart App design.

Figure below shows the code used to calculate the surface blasting design and mean fragment size after the user has entered the valid input parameters.

Discussion

The python surface blasting design and fragmentation calculators, that were developed, can successfully calculate the stemming length, burden, actual powder factor and the mean fragment size, provided the user inserts valid inputs. However, when invalid inputs are entered, the program displays an 'error' message which allows the user to re-enter the inputs. The code uses a function that allows the separation of different rock types which were used to determine the rock factor and technical powder factor, allowing modifications and improved accuracy throughout the code.

Future Work

1.